

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Isolated adjustable emitter supply

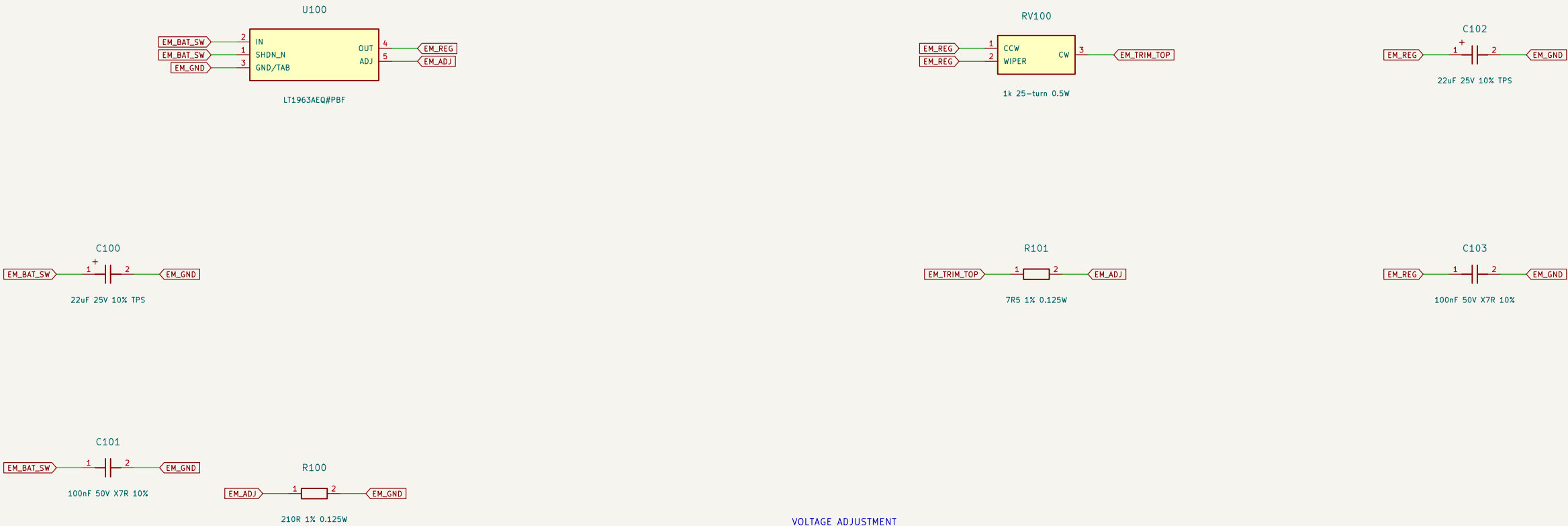
Three optically isolated emitter PWM channels

File: emitter_regulator.kicad_sch

File: emitter_pwm.kicad_sch

Sheet: / File: single_detector_usb_master.kicad_sch		
Title: Single detector rig – USB master power		
Size: A3	Date: 2026-09-09	Rev: R1 – DRAFT
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SUPPLY INTERFACE
EM_BAT_SW comes from the isolated positive master switch.
SHDN_N follows EM_BAT_SW; firmware does not bypass the master.
EM_GND remains separate from GND on every copper layer.
Input bypass connects directly IN-to-EM_GND; the original D1/C5 error is removed.



VOLTAGE ADJUSTMENT
 $V_{out} = 1.21 \cdot (1 + (7.5 + R_{trim}) / 210) + I_{adj} \cdot (7.5 + R_{trim})$.
Vout = 1.21*(1+(7.5+Rtrim)/210) + Iadj*(7.5+Rtrim).
Nominal electrical setpoint: 1.253..7.019V, limited by battery/dropout.
This preserves the original broad adjustment; it is NOT a safe heater preset.
Use the established measured emitter voltage and validated duty/frequency.
Mark the chosen setting; do not turn the pot to its end stop to calibrate.

THERMAL DESIGN
Three channels: 200mA each electrical design envelope.
600mA at every trim voltage is NOT thermally qualified.
 $P = (V_{in} - V_{out}) \cdot I_{out} + V_{in} \cdot I_{gnd}$. See research/emitter_stage.md.
TO-263 tab/pin3: broad EM_GND copper and thermal vias.
No solid ground plane may bridge the optical isolation boundary.

ENGINEERING WORK IN PROGRESS – NOT FOR FABRICATION

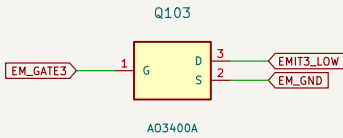
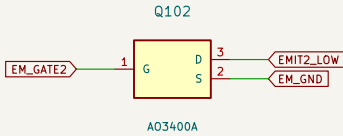
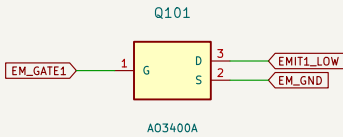
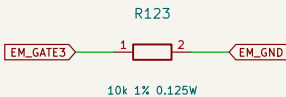
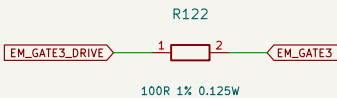
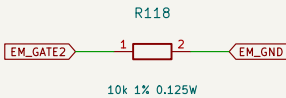
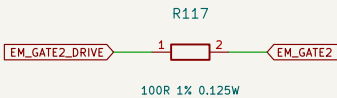
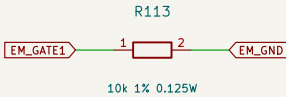
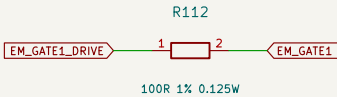
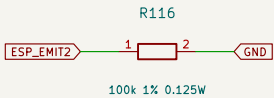
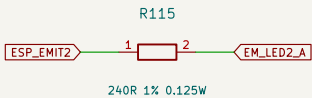
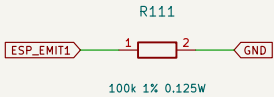
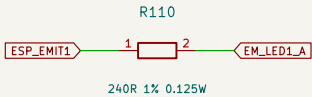
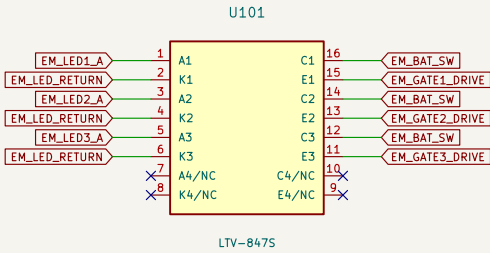
Eltec Instruments

Sheet: /Isolated adjustable emitter supply/
File: emitter_regulator.kicad_sch

Title: Isolated adjustable emitter supply

Size: A3	Date: 2026-09-09	Rev: R1 – DRAFT
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GPIO33 -> ESP_EMIT1 -> emitter1. GPIO25 -> emitter2. GPIO26 -> emitter3.
All outputs default off when control pins float. An asserted master permit alone does not pulse emitters.
Connect each heater positive to EM_REG; connect its return ONLY to its EMITn_LOW net.
Do not connect the switched heater return directly to EM_GND, which would bypass PWM.



GND = ESP32/USB side. EM_GND = isolated emitter battery side.
EM_LED_RETURN is sunk to GND only while the power stage permits operation.
MASTER_PERMIT therefore blocks all optical drive independently of the three GPIO commands.
Verify startup/shutdown, PWM timing, low setting gate enhancement, current and temperature.

ENGINEERING WORK IN PROGRESS – NOT FOR FABRICATION

Eltec Instruments

Sheet: /Three optically isolated emitter PWM channels/
File: emitter_pwm.kicad_sch

Title: Three optically isolated emitter PWM channels

Size: A3 Date: 2026-09-09

KiCad E.D.A. 10.0.6

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Id: 3/3